

- Vigas
- B_1 C_1
- $\frac{4EI}{L}$ $\frac{6I}{L} = \frac{EIc}{2h}$ $\frac{4(\frac{EIc}{2})}{2h} = \frac{EIc}{h}$ $\frac{2EI}{L} = 2 \cdot \frac{EIc}{2h} = \frac{EIc}{h}$
- $K_{BC} = \begin{bmatrix} \frac{EIc}{h} & \frac{EIc}{2h} \\ \frac{EIc}{2h} & \frac{EIc}{h} \end{bmatrix}$ $B_1 = \frac{EIc}{h^3}$
- $\begin{bmatrix} h^2 & \frac{h^2}{2} \\ \frac{h^2}{2} & h^2 \end{bmatrix}$ B_1 C_1

$$\frac{EI_c}{h^3} \begin{bmatrix} A_1 & A_2 & B_1 & B_2 \\ 4h^2 & -6h & 2h^2 & 6h \\ -6h & 12 & -6h & -12 \\ 2h^2 & -6h & 4h^2 & 6h \\ 6h & -12 & 6h & 12 \end{bmatrix} \begin{matrix} A_1 \\ A_2 \\ B_1 \\ B_2 \end{matrix}$$

$$K_{DC} = \frac{EI_c}{h^3} \begin{bmatrix} D_1 & D_2 & C_1 & C_2 \\ 4h^2 & -6h & 2h^2 & 6h \\ -6h & 12 & -6h & -12 \\ 2h^2 & -6h & 4h^2 & 6h \\ 6h & -12 & 6h & 12 \end{bmatrix} \begin{matrix} D_1 \\ D_2 \\ C_1 \\ C_2 \end{matrix}$$

$$\frac{EI_c}{h^3} \begin{bmatrix} B_1 & C_1 \\ h^2 & \frac{h^2}{2} \\ \frac{h^2}{2} & h^2 \end{bmatrix} \begin{matrix} B_1 \\ C_1 \end{matrix}$$

Matriz global de la estructura

$$\frac{EI_c}{h^3} \begin{bmatrix} A_1 & A_2 & B_1 & B_2 & C_1 & C_2 & D_1 & D_2 \\ 4h^2 & -6h & 2h^2 & 6h & 0 & 0 & 0 & 0 \\ -6h & 12 & -6h & -12 & 0 & 0 & 0 & 0 \\ 2h^2 & -6h & 5h^2 & 6h & h^2/2 & 0 & 0 & 0 \\ 6h & -12 & 6h & 12 & 0 & 0 & 0 & 0 \\ 0 & 0 & h^2/2 & 0 & 5h^2 & 6h & 2h^2 & -6h \\ 0 & 0 & 0 & 0 & 6h & 12 & 6h & -12 \\ 0 & 0 & 0 & 0 & 2h^2 & 6h & 4h^2 & -6h \\ 0 & 0 & 0 & 0 & -6h & -12 & -6h & 12 \end{bmatrix} \begin{matrix} A_1 \\ A_2 \\ B_1 \\ B_2 \\ C_1 \\ C_2 \\ D_1 \\ D_2 \end{matrix}$$

$B_2 = C_2$

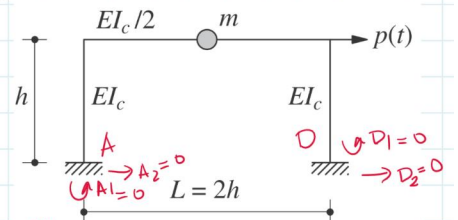
Matriz reducida

Suma B_2 y C_2 Columnas

$$\frac{EI_c}{h^3} \begin{bmatrix} B_2+C_2 & B_1 & C_1 \\ 6h & 5h^2 & h^2/2 \\ 12 & 6h & 0 \\ 6h & h^2/2 & 5h^2 \\ 12 & 0 & 6h \end{bmatrix} \begin{matrix} B_1 \\ B_2 \\ C_1 \\ C_2 \end{matrix}$$

Suma B_2 y C_2 Filas

$$\frac{EI_c}{h^3} \begin{bmatrix} B_2+C_2 & B_1 & C_1 \\ 24 & 6h & 6h \\ 6h & 5h^2 & \frac{h^3}{2} \\ 6h & h^2/2 & 5h^2 \end{bmatrix} \begin{matrix} B_2+C_2 \\ B_1 \\ C_1 \end{matrix}$$



Condensación matricial

$\approx$ masa Cero

$$K_C = K_{aa} - K_{ab} K_{bb}^{-1} K_{ba}$$

$$\begin{bmatrix} K_{aa} & K_{ab} \\ K_{ba} & K_{bb} \end{bmatrix} \begin{matrix} a \\ b \end{matrix}$$

$$\frac{EI_c}{h^3} \begin{bmatrix} B_2+C_2 & B_1 & C_1 \\ 24 & 6h & 6h \\ 6h & 5h^2 & \frac{h^3}{2} \\ 6h & h^2/2 & 5h^2 \end{bmatrix} \begin{matrix} B_2+C_2 \\ B_1 \\ C_1 \end{matrix}$$

$$K_{aa} = 24$$

$$K_{ab} = \begin{bmatrix} 6h & 6h \end{bmatrix}$$

$$K_{bb} = \begin{bmatrix} 5h^2 & h^3/2 \\ h^3/2 & 5h^2 \end{bmatrix}$$

$$K_{ba} = \begin{bmatrix} 6h \\ 6h \end{bmatrix}$$

$$K_C = K_{aa} - \underbrace{K_{ab} K_{bb}^{-1} K_{ba}}$$

$$K_{aa} = 24$$

$$K_{bb}^{-1} = \begin{bmatrix} \frac{20}{99 h^2} & -\frac{2}{99 h^2} \\ -\frac{2}{99 h^2} & \frac{20}{99 h^2} \end{bmatrix}$$

$$K_{bb}^{-1} * K_{ba} = \begin{bmatrix} \frac{20}{99 h^2} & -\frac{2}{99 h^2} \\ -\frac{2}{99 h^2} & \frac{20}{99 h^2} \end{bmatrix} * \begin{bmatrix} 6 h \\ 6 h \end{bmatrix} = \begin{bmatrix} \frac{12}{11 h} \\ \frac{12}{11 h} \end{bmatrix}$$

$$K_{ab} * (K_{bb}^{-1} * K_{ba}) = \begin{bmatrix} 6 h & 6 h \end{bmatrix} * \begin{bmatrix} \frac{12}{11 h} \\ \frac{12}{11 h} \end{bmatrix} = \frac{144}{11}$$

$$K_C = \frac{EI_C}{h^3} \left(24 - \frac{144}{11} \right) = \frac{120 EI_C}{11 h^3}$$

No amortiguada

$$m \ddot{u} + \frac{120 EI_C}{11 h^3} u = p(t)$$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{\frac{120 EI_C}{11 h^3}}{m}} = \sqrt{\frac{120 EI_C}{11 m h^3}}$$